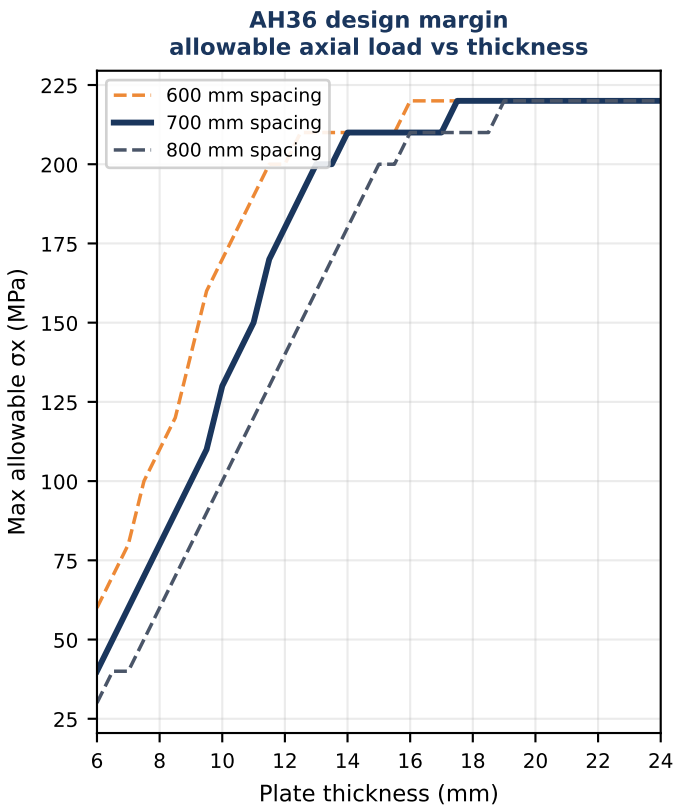
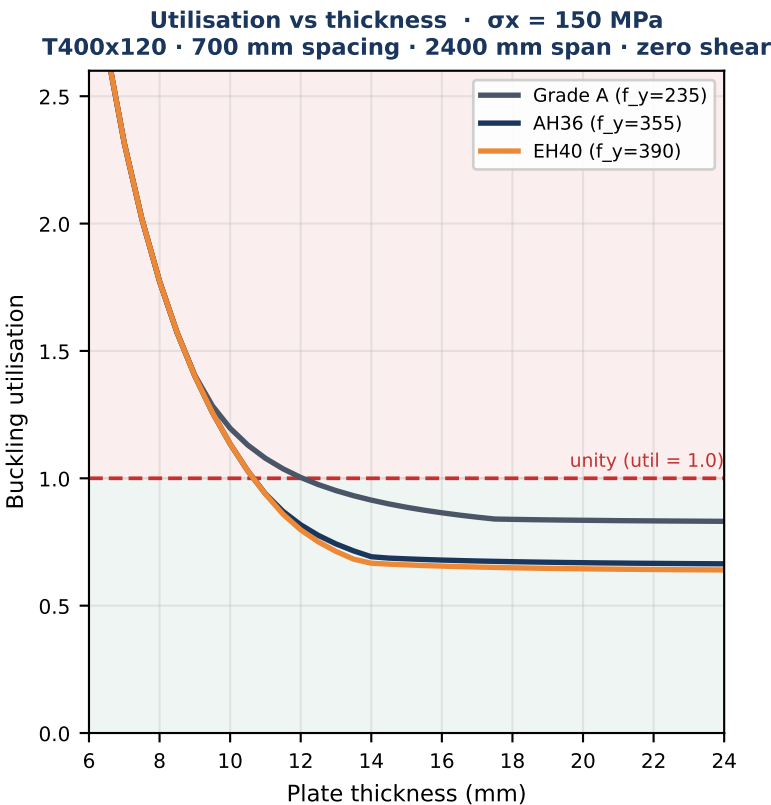


Ship Hull Stiffened-Panel Buckling — Design Brief

DNV-RP-C201 buckling assessment · marine grades: Grade A (235), AH36 (355), EH40 (390 MPa)

Validated solver — StiffenedPanelBucklingAnalyzer (golden-tested against DNV-RP-C201)



KEY FINDINGS

- Reference AH36 panel (700 mm spacing, T400x120, 2400 mm span, zero shear): minimum acceptable plate thickness for $\sigma_x = 150$ MPa is 11 mm (first thickness with utilisation ≤ 1.0).
- Grade sensitivity at $\sigma_x = 150$ MPa: Grade A needs ≈ 12.5 mm, AH36 ≈ 11 mm, EH40 ≈ 11 mm — higher grade lifts capacity and trims required plate.
- Governing failure modes at the reference point: plate-field / stiffener tripping. Overall column buckling never governs at ship frame spacings.
- Longer stiffener span and added in-plane shear both reduce the allowable axial load — see the AH36 margin curve for the spacing trade-off.

AH36 max allowable σ_x (MPa)
700 mm spacing · T400x120 · 2400 mm · $\tau=0$

t (mm)	Allowable σ_x
10	130
12	180
14	210
16	210
18	220